Ushtrime ND7c

1.2.97 *Në ç'lartësi nga sipërfaqja e Tokës nxitimi i rëndimit tokësor do të jetë g=1 m/s²?*

***Zgjidhja***

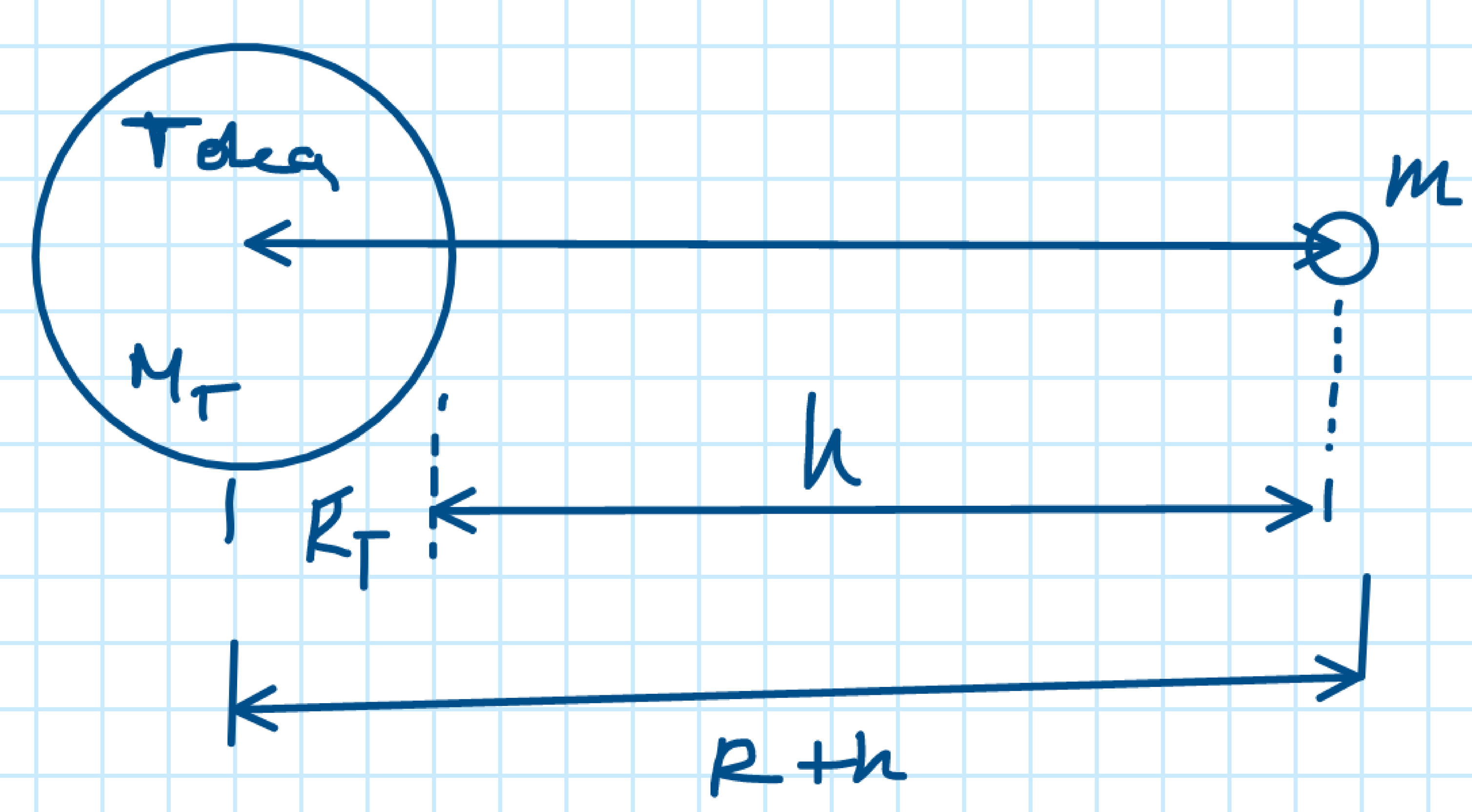
Të dhënat:

$h = ?$

$g = 1 \frac{m}{s^2}$

$R_T = 6.37 \cdot 10^6 m$

$g^{\theta} = 9,81 \frac{m}{s^2}$




$$\left.\begin{array}{l} F = \gamma \frac{m M_T}{R_T^2} \\ F = m g^{\theta} \end{array}\right\} \cancel{m} g^{\theta} = \gamma \frac{\cancel{m} M_T}{R_T^2}$$

$$\boxed{g^{\theta} = \gamma \frac{M_T}{R_T^2}}$$

$h = ? \longrightarrow g = 1 \frac{m}{s^2}$

$$\left.\begin{array}{l} F = \gamma \frac{m M_T}{(R_T+h)^2} \\ F = m g \end{array}\right\} \cancel{m} g = \gamma \frac{\cancel{m} M_T}{(R_T+h)^2}$$

$$\boxed{g = \gamma \frac{M_T}{(R_T+h)^2}}$$

$$\frac{g^{\theta}}{g} = \frac{\cancel{\gamma} \frac{\cancel{M_T}}{R_T^2}}{\cancel{\gamma} \frac{\cancel{M_T}}{(R_T+h)^2}} = \frac{(R_T+h)^2}{R_T^2} \Big/ \cdot R_T^2$$

$$\frac{g^{\theta}}{g} R_T^2 = (R_T+h)^2 \Big/ \sqrt{\phantom{x}}$$

$$R_T + h = \sqrt{\frac{g^{\theta}}{g} \cdot R_T^2}$$

$$R_T + h = R_T \sqrt{\frac{g^{\theta}}{g}}$$

$$h = R_T \sqrt{\frac{g^{\theta}}{g}} - R_T$$

$$h = R_T \left( \sqrt{\frac{g^{\theta}}{g}} - 1 \right)$$

$$h = 6.37 \cdot 10^6 m \left( \sqrt{\frac{9,81 \cancel{\frac{m}{s^2}}}{1 \cancel{\frac{m}{s^2}}}} - 1 \right)$$

$$h = 6.37 \cdot 10^6 m \, (3,13 - 1)$$

$$h = 6,37 \cdot 2,13 \cdot 10^6 m$$

$$h = 13,5681 \cdot 10^6 m$$

$$h = 13568,1 \cdot 10^3 m$$

$$\boxed{h = 13568,1 km}$$